Fiscal Year:	FY 2018	**Task Last Updated:**	FY 03/21/2018
PI Name:	Wood, Scott J. Ph.D.		
Project Title:	Vestibular-Evoked Myogenic Potentials in Microgravity		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: PI returned to NASA JSC in January 2017. PI was at Azusa Pacific University from August 2013 – January 2017; prior to August 2013, PI was at NASA JSC.		
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	12/11/2015	**End Date:**	10/01/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/2017 per HRP/JSC; ESA has descoped the project (Ed., 6/7/17)		
Key Personnel Changes/Previous PI:	Effective January, 2017, Dr. Scott Wood was no longer a Co-Investigator following his transition to NASA Johnson Space Center (JSC); Dr. Wood remained as advisor until 10/1/2017.		
COI Name (Institution):	Clement, Gilles Ph.D. (ESA PI: Lyon Neuroscience Research Center)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This study was proposed to study adaptive changes in otolith function using vestibular-evoked myogenic potentials (VEMPs) during and following long-duration spaceflight. Vestibular-evoked myogenic potentials are short latency vestibular reflexes recorded from neck or extraocular muscles in humans. The ocular VEMPs recorded from eye muscles reflect utricular-ocular reflex function, while cervical VEMPs recorded from contracted neck muscles reflect saccular-spinal reflex function. Given that there is an off-loading of the otolith organs in microgravity, we hypothesized that neurosensory adaptation will involve changes in sensitivity of otolith function as reflected by VEMP response amplitudes and thresholds. We also predicted that exposure to microgravity will reveal asymmetries in otolith function that vary with motion sickness severity in the astronaut population. Methods: We proposed to record VEMP measures on 12 long-duration crewmembers before (3 preflight sessions), during (up to 6 inflight sessions occurring approximately every month), and following spaceflight (3 post-flight sessions within 1, 4, and 8 days of return). Each proposed session included cVEMP and oVEMP threshold responses to examine changes in otolith sensitivity, as well as bilateral responses to examine response asymmetries. We also proposed to correlate these VEMP measures to preflight utricular function tests using eye movements and perceptual responses to unilateral centrifugation. VEMPs can be evoked by a variety of vestibular stimuli, such as air-conducted (AC) sound, bone-conducted (BC) vibration, and bone-conducted head taps. The sound stimuli range from clicks to tone bursts over different frequencies. The vibration and head taps have been applied to the forehead or lateral sides of the head closer to the vestibular end organs. Following selection of our flight proposal, we conducted a ground control study on normative subjects to characterize the baseline VEMP responses in a healthy population with different vestibular stimuli to assess the feasibility of using each during the flight study. Risk Characterization, Quantification\Evidence: This task was intended to contribute to gap closure by providing a better understanding of the long-term adaptation of the otolith system to the conditions of spaceflight and by examining the contributions of this adaptation to motion sickness severity and sensorimotor performance deficits. Flight/Ground/Analog and resources: This study was initially selected for feasibility assessment by the European Space Agency and not carried in the U.S. International Space Station (ISS) utilization plans. It was subsequently deselected from ISS by ESA and redirected for parabolic flight without NASA involvement.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The otoliths in the vestibular system are the one sensory organ that directly transduce both magnitude and direction of gravitational acceleration. Therefore, it is somewhat surprising that there have been no direct measures of otolith function during long duration spaceflight. We hypothesize that the oculomotor and perceptual disturbances experienced early inflight and perceptual and motor coordination problems experienced postflight both reflect adaptive changes in how inertial cues from the otolith system are integrated with other sensory information. Although the effects are most profound shortly following g-transitions, the fact that the postflight symptom severity worsens with flight duration suggests that there are longer adaptation time constants whereby subtle changes occur over weeks-to-months. We predict an initial reduction in otolith sensitivity (with increased asymmetry) early inflight. One hypothesis is that this will be followed by return to baseline, with increased otolith sensitivity during later phases of the mission and early postflight. An alternate hypothesis is that the otolith sensitivity will continue to be reduced throughout the mission, with increased weighting of alternate sensory systems, e.g., proprioception and vision. This study was designed to directly measure how otolith sensitivity changes over long-duration spaceflight to inform future sensorimotor countermeasure development. The VEMP techniques utilized in this research study have the potential for differential diagnosis of utricular and saccular reflex pathways based on neurophysiological studies. While there have been numerous reports of the diagnostic performance of VEMPs to detect various vestibular disorders, the nature of the evoked myogenic potentials requires careful control to minimize false negative results. The results of this ground study suggested that measurements of head acceleration should be considered to control for response variability when using hand-held administered application of bone-conducted stimuli.
Task Progress:	As part of the flight study preparation, an extensive literature review was conducted on the variations in VEMP methodology across laboratories (Wood and Clément, in preparation). The goal of this review is to examine different thresholding methods that might help establish variations in otolith sensitivity. We had proposed a novel head tapper technique but were advised as part of the merit review process to consider alternate acoustic and bone conduction methods. These techniques have been used for both detecting asymmetries in function (unilateral pathology) and changes in threshold due to hypersensitivity (semicircular canal dehiscence). However, our proposal to use VEMP methods to track changes in otolith sensitivity is quite unique. Based on our literature review, the main objective of our initial ground control study was to compare responses from two bone-conducted (BC) methods – vibration through a mini-shaker and step impulses delivered with an automated head tapper. We were interested in determining which method would provide the most reliable measure of threshold sensitivity for both oVEMP and cVEMP responses. Similar to our flight study, we also planned to obtain utricular function measures using unilateral eccentric rotation similar to what had been conducted pre and post-Shuttle (Otolith study). Custom LabView software was developed to implement the head tapper and bone conduction protocols. The head tapper software was adapted from an existing system that was implemented for Dr. Seidler's Neuromapping study. The bone conduction protocol was adapted from Welgampola and colleagues using a Brüel & Kjær mini-shaker with a custom fiberglass rod interface. Since tapper and bone conduction techniques rely on the operator manually applying a probe and one objective of our study was to incorporate threshold stimuli, we also incorporated a bite-board attached sensor to directly measure skull head acceleration at the various stimulus levels. Fifteen subjects participated in this experiment. The cVEMP response was recorded by electrodes placed over tensed sternocleidomastoid muscles. The oVEMP response was recorded by electrodes placed just beneath the eyes over the inferior oblique and inferior rectus extraocular muscles. Bilateral cVEMP and oVEMP responses were elicited by a handheld head tapper and a mini-shaker, both set for delivering head acceleration at four different levels. Head acceleration and the resulting VEMP amplitude were measured for each level. In addition, measurements of the subjective visual vertical (SVV) and verbal reports of tilt perception were obtained during unilateral centrifugation involving 10 deg equivalent roll tilt.

	The head tapper was effective at eliciting both oVEMP and cVEMP in 67-90% of the trials with head accelerations ranging from 0.17 to 0.24 g. However, there was a large variability in the head accelerations recorded for each stimulus level. By contrast, the mini-shaker was effective at inducing discrete head accelerations, but at the levels recorded in this study (<0.12 g) produced oVEMP and cVEMP in only 10-30% of the trials. Measurements of subjective vertical during unilateral centrifugation showed normal range of responses; however, the inconsistency of eliciting VEMP responses limited comparisons to the SVV. While the cVEMPs and oVEMP responses were more robust with the head tapper than the mini-shaker, it is not clear how much this difference was attributable to the stimulus profile (step impulse versus tone burst), magnitude of the head accelerations imparted, and/or other factors related to the hand-held application of the stimuli (e.g., direction of the resultant force relative to the otolith organs). Additional development would be required to validate either method to effectively measure changes in otolith gain during spaceflight, especially at threshold levels. While not used in this ground-based study, the auditory tone method offers advantages of a more consistent application of stimulus levels over a wide range (threshold to maximum response), as well as increased feasibility of self-administered testing to save crew time. There are well-known guidelines for controlling VEMP responses, namely neck muscle contraction for the cVEMP and gaze direction for the oVEMP. For studies using hand-held administered application of bone-conducted stimuli, the results of this ground study suggest that measurements of head acceleration should also be considered to control for response variability.
Bibliography Type:	Description: (Last Updated: 02/20/2026)
Abstracts for Journals and Proceedings	Campbell D, Clément G, Reschke MF, Bloomberg JJ. "Evaluating otolith function during spaceflight using vestibular evoked myogenic potentials." 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
Articles in Peer-reviewed Journals	Wood SJ, Clément GC. "Examining changes in otolith sensitivity during long-duration spaceflight using Vestibular Evoked Myogenic Potentials." BMC Neurology (in preparation for submission as of December 2018). , Dec-2018